

Softmax Regression on Digits Dataset

Abstract

This paper evaluates a simple softmax regression model on the 8x8 digits dataset.

Experimental setup

We used the 8x8 digits dataset (1,797 images) with an 80/20 train/test split.

The model was trained using SGD.

learning rate = 0.5

epochs = 20

batch size = 32

seeds 0 to 4

Table 1

Hyper-parameters are as listed above.

Results

We achieved a test accuracy of 0.956 ± 0.002 (mean $\pm$ std over 5 seeds).